SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA5116
COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: **Total Marks: 50**

Annexure A

Analytical Problem Solving

Code of Conduct:

IM.SADANA[192524386] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Sadhana

Cryptography And Network Security

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Date :

2) Block Cipher Mode Evaluation

Given :

Plaintext blocks : $P = [1010, 1010, 1111, 1010]$

Key : $K = 1100$

Initialization Vector : $IV = 0000$.

Assume the encryption function is $E_K(X) = X \oplus K$ (XOR operation).

Encryption using ECB and CBC modes.

ECB Mode [Electronic codebook]

Formula :

$$C_i = P_i \oplus K$$

Block	plaintext	Key	ciphertext
1	1010	1100	0110
2	1010	1100	0110
3	1111	1100	0011
4	1010	1100	0110.

ECB ciphertext : 0110, 0110, 0011, 0110

CBC Mode [Cipher Block chaining]

formula :

$C_i = E_K(P_i \oplus C_{i-1})$, where $C_0 = IV$

Block	calculation	ciphertext
1	1010 ⊕ 0000 ⊕ 1100	0110
2	1010 ⊕ 0110 ⊕ 1100	0000
3	1111 ⊕ 0000 ⊕ 1100	0011
4	1010 ⊕ 0011 ⊕ 1100	0101

CBC Ciphertext : 0110 , 0000 , 0011 , 0101

Compare the ciphertext outputs

ECB Mode	CBC Mode
Repeated plaintext gives repeated ciphertext	Repeated plaintext can give different ciphertext
Blocks are encrypted independently.	Each block depends on the previous block
Simple and fast.	Requires chaining between blocks

p which mode hides patterns better?

CBC mode hides patterns than ECB mode because each plaintext block is combined with the previous ciphertext block before encryption. Therefore, Identical plaintext blocks can produce different ciphertext blocks.

DES vs 3-DES vs Blowfish performance study

Given:

* DES: Key size = 56 bits ; time = 2ms/block
* 3-DES : Key size = 168 bits; time = 6ms/block
* Blowfish: Key size = 128 bits ; time = 1ms/block
* Total data = 1000 blocks.

Calculate to encryption time.

Formula:

Total Time = time per Block x num

locks.

Calculate total encryption time.

rmula :

Total Time = Time per Block × Number of Blocks

Algorithm	Time per Block	Total Blocks	Total Time
DES	2ms	1000	2000ms = 2 seconds
3-DES	6ms	1000	6000 ms = 6 seconds
Blowfish	1ms	1000	1000 ms = 1 second

). Analyze trade-offs between security and performance.

- DES: Fast, but its 56-bit key is too short for modern security requirements.
- 3-DES: provides stronger protection than DES but slower because it performs encryption three times
- Blowfish provides a longer key size and the fastest performance among the given options. However, its 64-bit block size makes it less suitable for encrypting large amount of modern d

ecommend the best algorithm for secure n
e communication.

ecommendation: Blowfish among the three given lgorithm, because it has the shortest encryption ime (1 second for 1000 blocks) and a larger key ze than DES.

owever, for a real-world secure communication stem, AES-128 or AES-256 is generally referred because it offers modern security and cient performance. Blowfish is an older algorithm d is not the best choice for new systems.